SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	T Bhanu Teja	**Reg. No.:**	192472259
Section / Batch:	2024	**Date:**	07-08-26

Code of Conduct

I _____T Bhanu Teja(192472259)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Bhanu Teja__

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as "analyzing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as "govern", "government", and "governance". The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4.A document classification system processes input words such as "activate", "activation", and "reactivation" to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation.
Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as "create", "creates", and "creating" to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

07-08-26

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Q1. Morphological processing for "analyzing", "analysis", and "analytical".

Aim: To design a morphological processing approach that decomposes the given words into root and affixes, classifies them as inflectional or derivational, and normalizes them to a common base form for efficient document indexing.

Morphological Decomposition:

Word	Root	Affix	Type
analyzing	analyze	-ing	Inflectional
analysis	analyze	-sis	Derivational
analytical	analyze	-ical	Derivational.

Normalization: All three words are converted to common base form

- Analyzing
- Analysis
- Analytical

↓

Analyze

→ This allows the search engine to retrieve all related documents even if different word forms are used.

Final output

Word	Morphological structure	Normalized
analyzing	analyze + ing	analyze
analysis	analyze + sis	analyze
analytical	analyze + ical	analyze

Conclusion :- The morphological professor identifies the root word "analyze", distinguishes inflectional and derivational forms, and normalizes all variants into a single representation.

Q2. morphological Parsing for "disagree", "agreement", and "agreeable".

Aim :- To design a morphological parsing strategy that identifies prefixes, suffixes, and base forms while preserving semantic meaning for sentiment analysis.

Morphological Decomposition :-

Word	Prefix	Root	Suffix	Type
disagree	dis-	agree	—	Derivational
agreement	—	agree	-ment	Derivational
agreeable	—	agree	-able	Derivational.

Morphological Parsing :-

disagree → dis + agree

agreement → agree + ment

agreeable → agree + able.

Normalized Root :-

disagree
agreement
agreeable
↓
agree

Conclusion :-

The Parser succesfully identifes the common root "agree" and analyzes the ffect of prefixes and suffixes. This helps sentiment analysis system understand related words while preserving their semantic differences.

Q3. Morphology-Based Normalization for "govern", "goverment", and "governance".

Aim: To normalize related words into a common base form for topic modling and document clustering.

Morphological Decomposition :-

Word	Root	Suffix	Type
Govern	Govern	—	Base word
goverment	govern	-ment	Derivational
governance	govern	-ance	Derivational.

Classification :-

* Govern is the base verb.
* Goverment is a noun formed by adding -ment.
* Governance is a noun formed by adding -ance.

Both goverment and governance are derivational forms because they change the word class from verb to noun

Final output :-

Word	Morphological Structure	Normalization
Govern	Govern	govern
Goverment	Govern + ment	govern
Governance	Govern + ance	govern

Conclusion :- The Normalization process maps all words to the common root "govern", improving document clustering and retrival by treating related words as one concept.

Q4. Morphological Analysis for "activate", "activation", and "reactivation".

Aim :- To analyze derivational layers, identify prefixes and suffixes, and normalize all forms to a common root.

Morphological Decomposition :-

Word	Prefix	Root	Suffix	Type
activate	—	active	-ate	Derivatoral
activation	—	activate	-ion	Derivational
reactivation	re-	activate	-ion	Derivational

Classification :-

→ "Activate" is formed by adding -ate to create a verb.

* activation adds -ion, changing the verb into a noun.
* reactivation adds the prefix re, -meaning "again", and the suffix -ion.

Morphological Parsing :-

activate → active + ate
activation → activate + ion
reactivation → re + activate + ion.

Normalized Root :-

activate
activation
reactivation
↓
activate.

Final output :-

word	morphological structure	Root
activate	active + ate	activate
activation	activate + tion	activate
reactivation	~~re~~ re + activate + tion	activate.

Conclusion :-

The morphological analyzer extracts the common root "activate" while identifying derivation layers and semantic changes, ensuring consistent indexing & retrival.

Q5. Morphology - Based Normalized for "create", "creats", and "creating"

Aim:- To identify inflectional suffixes, extract the common root, and normalize all grammatical forms for document retrival.

Morphological Decomposition:

Word	Root	Suffix	Type
Create	create	—	Base word
creates	create	-s	Inflectional
creating	create	-ing	Inflectional.

Classification:

* creates uses the suffix -s to indicate the third person singular present tense.
* Creating use the suffix -ing to indicate the present participle.

Final output:

Word	Morphological Structure	Normalized form
Create	Create	Create
creates	Create +s	Create
Creating	Create +ing	create.

Conclusion:

The normalization process removes inflectional suffixes and converts all word forms to the common root "create". This improve indexing, search efficiency, and retrival accuracy by treating all grammatical variants as a single term.